

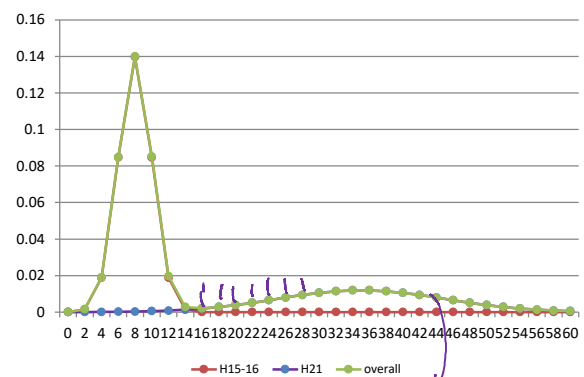
Agenda

- Last time
 - Expected values of sums of random variables
 - Joint distributions
 - Conditional probability
- Today
 - Independence i.i.d
 - Covariance P.C.A

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Mixture

- Discrete:
 - $Z = 0$: Student residence is H7-8-21
 - $Z = 1$: Student residence is H15-16
- Continuous: X : travel time to (New) SAC (or 9Tea7 Shach)
- $f_{X,Z}(x, z) = P(Z = z)f_{X|Z}(x | z)$



How do we interpret this graph?

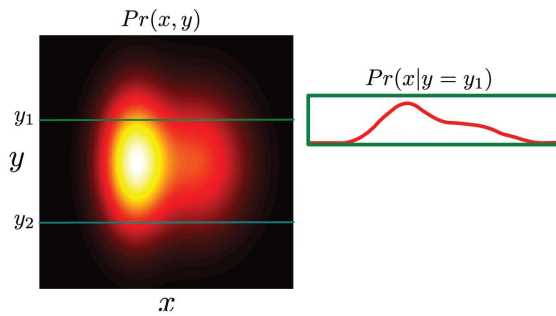
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Independence

- If two variables X and Y are independent, then variable X tells us nothing about variable Y (and vice-versa)

$$Pr(x|y) = Pr(x)$$

$$Pr(y|x) = Pr(y)$$



What will the second distribution look like at $y = y_2$?

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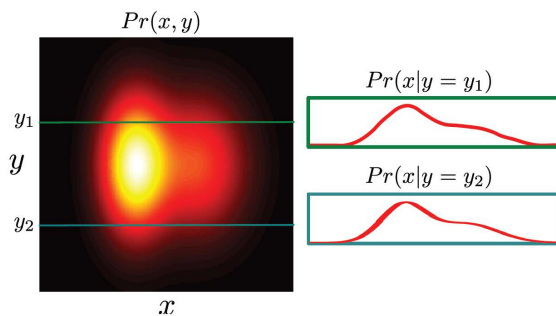
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Second distribution at $y = y_2$ is identical

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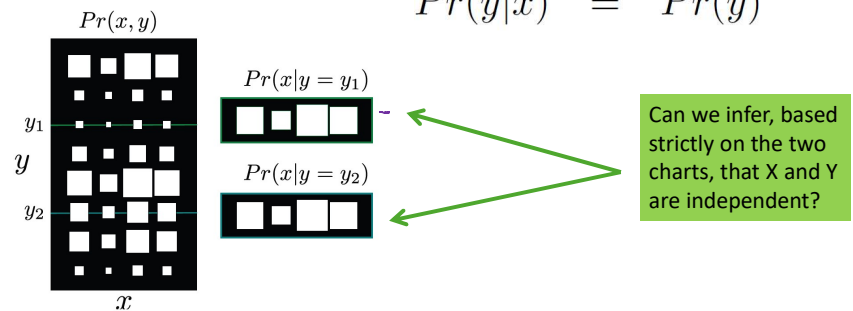
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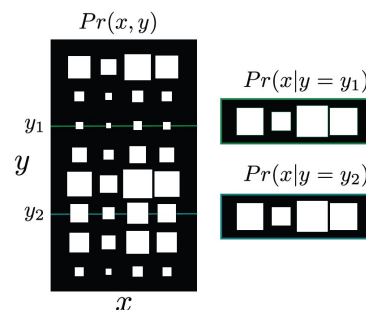
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Independence

- When variables are independent, the joint factorizes into a product of the marginals:

$$Pr(x, y) = Pr(x|y)Pr(y)$$

$$= Pr(x)Pr(y)$$



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Independence

- The joint pdf is the product of the marginal pdfs.
- For continuous independent random variables the joint cdf also factorizes

$$\forall x, \forall y, \quad f_{XY}(x, y) = f_X(x) f_Y(y)$$

$$F_{XY}(x, y) = F_X(x) F_Y(y)$$

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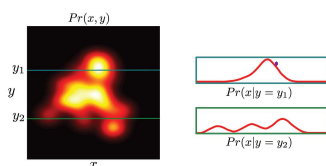
Quick Summary



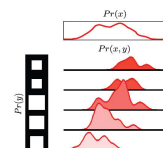
$$F_{XY}(a, b) = \int_{-\infty}^a \int_{-\infty}^b f_{XY}(x, y) dx dy$$

$$f_{XY}(a, b) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x, y) \big|_{x=a, y=b}$$

- Rules of probability are compact and simple
- Concepts of joint, marginal, and conditional probability, and expectation underpin most of the machine learning models



Conditional
probability



Marginalization



Probability = area
of density surface

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Expectation Rules

Must know

- Expected value of a constant is the constant

$$E[f[X]] = \int f[x]Pr(X = x)dx$$

$$E[\kappa] = \kappa$$

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Expectation Rules

Must know

- Expected value of constant times function is constant times expected value of function

$$E[f[X]] = \int f[x]Pr(X = x)dx$$

$$E[kf[x]] = kE[f[x]]$$

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Expectation Rules

Must know

- Expectation of sum of functions is sum of expectation of functions

$$E[f[X]] = \int f[x]Pr(X = x)dx$$

$$E[f[x] + g[x]] = E[f[x]] + E[g[x]]$$

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Expectation Rules

Must know

- Expectation of product of functions in variables x and y is product of expectations of functions if x and y are independent

$$E[f[X]] = \int f[x]Pr(X = x)dx$$

$$E[f[x]g[y]] = E[f[x]]E[g[y]] \quad \text{if } x, y \text{ independent}$$

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Conditional Probability and Independence

$$Pr(x, y) = Pr(x|y)Pr(y)$$

$$\begin{aligned} Pr(w, x, y, z) &= Pr(w, x, y|z)Pr(z) \\ &= Pr(w, x|y, z)Pr(y|z)Pr(z) \\ &= Pr(w|x, y, z)Pr(x|y, z)Pr(y|z)Pr(z) \end{aligned}$$

How will this look like if the variables are independent?

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Mutual independence

This definition for n variables is hard to comprehend, see example next

- n continuous random variables are mutually independent iff
 - for every finite subset of indices $i_1, i_2, \dots, i_k$
 - And every choice of real numbers $x_i, \dots, x_k$
 - $P(X_{i_1} \leq x_1, \dots, X_{i_k} \leq x_k) = \prod_{j=1}^k P(X_{i_j} \leq x_j)$
- Pairwise independence is the much weaker condition when $k=2$

$$\forall x_1, x_2, \dots, x_n, \quad f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

$$\forall (x_i, x_j), 1 \leq i \leq n, 1 \leq j \leq n, i \neq j: \quad f_{X_i, X_j}(x_i, x_j) = f_{X_i}(x_i)f_{X_j}(x_j)$$

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Example

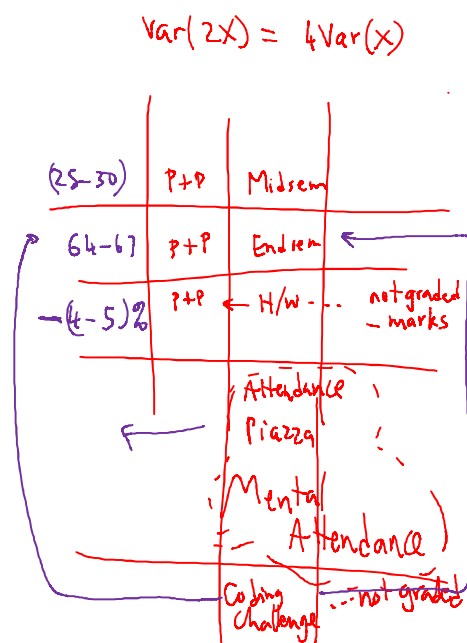
- Take the sample space $\{1, 2, 3, 4\}$, each singleton equally likely.
 - $P(\{1\}) = \frac{1}{4}$
- Let $A = \{1, 2\}$, i.e., either the event $\{1\}$ or the event $\{2\}$
 - $P(A) = \frac{1}{2}$
 - Similarly, $B = \{1, 3\}$, $C = \{1, 4\}$ and $P(A) = P(B) = P(C) = \frac{1}{2}$.
- Consider the intersection event $AB = \{1\}$. $P(AB) = \frac{1}{4}$
 - A and B are considered independent since $\frac{1}{2} * \frac{1}{2} = \frac{1}{4}$
- However, the intersection event ABC
 - Has $P(ABC) = P(\{1\}) = \frac{1}{4} \neq P(A)P(B)P(C) = \frac{1}{8}$, so A, B, C are not mutually independent.

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Evaluation

- “coding project”
 - Demonstrate Creativity ○
 - Demonstrate visualization tools
 - More details later
 - Groups of 4
- Percentages are approximate
 - Your cooperation is critical



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